

SCORANT WEBSITE CONTENT FOR EACH POC BUILT.

AI Knowledge Studio (ASR Studio)

AI Knowledge Studio (ASR Studio) is an intelligence platform that transforms raw meeting media into structured outputs for review, search, and reporting. The application supports audio files, video files, and document uploads. Images and real time transcriptions then routes each request through a query-first agent workflow to produce transcripts, speaker-separated conversations, summaries, Q&A responses, translations, and export-ready meeting artifacts.

End-to-End Processing Flow

The primary workflow begins when a user determines intent and content type, then triggers the appropriate tool. The user uploads a media or document file and submits a natural-language instruction through the agent bar. The backend then processes the request.

The platform implements Role-Based Access Control (RBAC) with department-wise permissions.

Agent-Oriented Interaction Model

The platform runs in two operating modes:

1. Plain Chat Mode

When no transcript or document context is active, the assistant behaves like a normal conversational chatbot with short-term session memory.

2. Contextual Agent Mode

When a transcript or document context is active (or a file is uploaded), the system routes to task-specific tools such as:

- media processing
- transcript Q&A
- document Q&A
- transcript summarization
- translation
- semantic search
- text-to-speech

Document Intelligence Chatbot

Overview

The Document Intelligence Chatbot transforms static organizational documents into an interactive knowledge platform where users can upload files and ask questions using natural language. The platform supports multiple formats including PDF, DOCX, PPT, Excel, TXT, scanned images, and image-based documents.

The system also supports multilingual documents by detecting the language of the extracted text and translating supported non-English content into English before indexing. This allows users to query documents consistently across different languages.

The platform implements Role-Based Access Control (RBAC) with department-wise permissions.

Response Generation

When a user submits a question:

- The query is converted into an embedding.
- The most relevant document chunks are retrieved from FAISS.
- Retrieved content is ranked by semantic relevance.
- The LLM generates an answer strictly from the retrieved evidence.
- Responses include source references such as page numbers or section names.

The platform also supports text-to-speech, allowing users to listen to generated responses.

Key Features

- Multi-format document support
- OCR for scanned files
- Multilingual document translation
- Semantic search using FAISS
- MongoDB metadata storage
- Role-Based Access Control
- Source-grounded responses
- Text-to-speech support
- Department-wise access management

Audio Intelligence Pipeline

Overview

The Audio Intelligence Pipeline converts spoken audio into structured and actionable outputs through a unified processing workflow. Users upload an audio file and select a processing mode such as Smart Summary or Language Translation.

Once uploaded, the audio is stored and assigned a unique request ID to ensure traceable and conflict-free processing.

The audio is then passed through a speech recognition stage where spoken content is converted into a text transcript. This transcript becomes the base content for all downstream processing.

Processing Modes

1. Smart Summary Mode

- The transcript is condensed into a concise summary.
- Important discussion points and key information are highlighted.
- The summary is designed for quick review and understanding.

2. Language Translation Mode

- The transcript is translated into the selected target language.
- Multiple output languages can be supported.
- The translated content preserves the meaning of the original audio.

3. Output Generation

- The processed text output is converted back into speech using text-to-speech.
- A playable audio response is generated.
- The backend returns:

Key Features

- Audio-to-text transcription
- Smart summarization
- Multi-language translation
- Text-to-speech synthesis
- Unique request tracking
- Combined text and audio output
- Simple and user-friendly workflow

Face Flow

Overview

Face Flow POC is a fully offline, on-premise facial recognition system developed for automated employee attendance management. The system uses deep learning-based face embeddings to identify registered employees in real time and record attendance with timestamps.

It operates entirely within a secure local infrastructure with no internet dependency, no external API calls, and complete data privacy. The platform uses DeepFace with the ArcFace model for facial recognition, OpenCV for face detection, and TensorFlow or PyTorch for model execution.

Attendance Logging

- If a match is found, the employee is marked as present.
- Each employee is recorded only once per day.

Unregistered User Handling

- If no facial match is found, the user is marked as unregistered.
- A unique label is assigned for tracking.
- Unknown users can be reviewed later by the admin.

Key Features

- Fully offline deployment
- Real-time face recognition
- On-premise attendance management
- DeepFace with ArcFace model
- OpenCV-based face detection
- CSV attendance logging
- Admin-controlled retraining and export
- Privacy-compliant biometric processing
- Secure local data storage

Object Detection Using YOLO

Overview

The object detection system is designed to identify and localize humans and weapons within images or video streams using a real-time deep learning detection framework. The model is trained on a specialized dataset to recognize multiple weapon types along with human presence in the scene.

By analyzing visual patterns, shapes, and contextual features, the system detects objects of interest and places bounding boxes around them with class labels and confidence scores.

This enables the model to identify both individuals and potential threats in a given environment.

Threat Monitoring

- The system can identify the presence of weapons near individuals.
- High-confidence detections can be flagged for further review.
- Alerts can be generated for potentially dangerous situations.
- Detection logs can be stored with timestamps and frame references.

Key Features

- Real-time object detection
- Human and weapon identification
- YOLO-based deep learning model
- Bounding box visualization
- Confidence score generation
- Video and image support
- Fast and efficient inference
- Security and surveillance use cases
- Threat monitoring and alert generation
- Timestamped detection logging

Change Detection Using Transformers

Overview

The change detection system is designed to identify and analyze differences between geospatial or satellite images captured at different time intervals. The model uses a transformer-based deep learning architecture to understand spatial relationships and contextual patterns across large image regions.

By comparing two images of the same location taken at different times, the system identifies variations in pixel density, textures, and structural patterns to detect areas where meaningful changes have occurred.

The system can capture both minor and major changes such as land development, infrastructure growth, environmental impact, and natural events.

Detection Workflow

- Two satellite or geospatial images of the same location are provided.
- Images are aligned and preprocessed for comparison.
- The transformer model extracts high-level visual features from both images.
- Pixel-level comparison is performed between the two image sets.
- Feature alignment and density mapping are used to identify changed regions.
- A change map is generated to highlight areas where differences are detected.

Output and Visualization

- The system highlights regions with significant changes.
- Change maps provide a visual representation of detected differences.
- Outputs can be used to compare before-and-after scenarios.
- Results can be stored with image timestamps and location references.

Key Features

- Satellite and geospatial image comparison
- Pixel-level change detection
- Feature extraction and alignment
- Change map generation
- Detection of land and infrastructure changes
- Environmental and disaster monitoring
- Large-scale area analysis
- Timestamped visual comparison
- Support for urban planning and geospatial intelligence